SkillNavora

Supplement A

Data Engineering, warehousing, pipelines and cloud

Data engineering fundamentals, warehousing and lakehouses, orchestration, data quality, cloud, Fabric and Databricks, containers and data architecture.

Part of the AI for Data Engineers (Data Science & AI) programme.

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Edition: November 2026. Official links point to the provider's own site. The sandbox used to write this could not open them, so check each link before you share it.

Supplement A: Data Engineering, Warehousing, Pipelines and Cloud

Each topic explains what it is, why it matters, where it is used, how it works, a worked example, a tool, an exercise, common mistakes and an interview question.

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A1 Data Engineering Fundamentals

A1.1

What data engineering is and the roles

WHAT IS IT?

> Data engineering builds and runs the systems that move, store and prepare data so analysts, data scientists and AI applications can use it reliably.

WHY IS IT IMPORTANT?

> Without reliable pipelines, dashboards show wrong numbers and models train on stale data. Engineers own that reliability.

WHERE IS IT USED IN INDUSTRY?

Banks, retailers, SaaS companies and AI product teams all employ data engineers to feed their reporting, ML and AI features.

HOW DOES IT WORK?

- Sources (apps, databases, APIs, files) produce raw data.
- Ingestion copies it into a landing zone.
- Transformation cleans and models it.
- Serving gives analysts and models trusted tables.

PRACTICAL EXAMPLE

> A sales app writes orders to a database; a nightly pipeline copies them to a warehouse and builds a daily revenue table that the Power BI dashboard reads.

TOOL OR SOFTWARE

Any SQL client and Python

HANDS-ON EXERCISE

Draw the path of one order from the app to the Power BI dashboard, naming each step and the owner.

COMMON MISTAKES

Treating a data engineer as a 'report builder'. Their work is the reliability and repeatability of the flow.

INTERVIEW QUESTION

What is the difference between a data analyst, a data scientist and a data engineer?

OFFICIAL AND FURTHER READING

- Apache Airflow docs (official)
- dbt docs (official)

A1.2

Data types and file formats: CSV, JSON, XML, Parquet, Avro, ORC

WHAT IS IT?

A file format is the way data is written to disk. Row formats (CSV, JSON, Avro) store records one after another; column formats (Parquet, ORC) store each column together.

WHY IS IT IMPORTANT?

Column formats read faster and smaller for analytics, and compression cuts storage cost. Choosing the wrong format is a common cause of slow, expensive pipelines.

WHERE IS IT USED IN INDUSTRY?

Data lakes and warehouses commonly store Parquet; Kafka pipelines often use Avro with a schema registry.

HOW DOES IT WORK?

- CSV: plain text, no types, easy to read.
- JSON: nested, flexible, larger.
- Parquet and ORC: columnar, typed, compressed, best for analytics.
- Avro: row format with a schema, common for streams.
- Compression (Snappy, gzip, zstd) trades CPU for size.

PRACTICAL EXAMPLE

A 2 GB CSV of transactions becomes about 250 MB as Parquet, and a query summing one column reads only that column.

TOOL OR SOFTWARE

pandas, DuckDB

HANDS-ON EXERCISE

Convert a CSV of 1 million rows to Parquet and compare file size and the time to sum one column.

COMMON MISTAKES

Storing everything as CSV because it is familiar, then wondering why queries are slow.

INTERVIEW QUESTION

Why is Parquet better than CSV for analytics?

OFFICIAL AND FURTHER READING

- Apache Parquet documentation (official)
- pandas IO tools (official)

A1.3

Ingestion: batch, streaming, API, file, database and CDC

WHAT IS IT?

> Ingestion is the step that brings data from sources into your platform. Batch loads chunks on a schedule; streaming sends events as they happen; CDC (change data capture) reads database changes from the transaction log.

WHY IS IT IMPORTANT?

> The ingestion method decides freshness, cost and risk of missed or duplicated records.

WHERE IS IT USED IN INDUSTRY?

Fraud teams use streaming; finance reporting usually uses daily batch; inventory systems often use CDC from the operational database.

HOW DOES IT WORK?

- Batch: scheduled pull of new or changed rows (often by an updated_at column).
- Streaming: events published to a topic and consumed continuously.
- API: paginated pulls with rate limits and tokens.
- CDC: reads inserts, updates and deletes from the database log.

PRACTICAL EXAMPLE

> An API returns orders in pages of 100; the pipeline stores the last updated timestamp and asks only for newer orders next run.

TOOL OR SOFTWARE

Python requests, Airbyte (open source connectors), Debezium for CDC

HANDS-ON EXERCISE

Write a script that pulls a paginated public API and stores only new records using a stored cursor.

COMMON MISTAKES

Pulling the whole table every night when only changes are needed.

INTERVIEW QUESTION

What is the difference between batch and streaming ingestion?

OFFICIAL AND FURTHER READING

- Debezium documentation (official)
- Airbyte docs (official)

A1.4

ETL versus ELT

WHAT IS IT?

> ETL extracts, transforms outside the target, then loads. ELT extracts and loads raw data first, then transforms it inside the warehouse.

WHY IS IT IMPORTANT?

> ELT is common on cloud warehouses that have cheap, fast compute, and keeps raw data for re-processing. ETL still fits when data must be cleaned or masked before it lands.

WHERE IS IT USED IN INDUSTRY?

Most modern analytics teams use ELT with dbt; regulated teams may use ETL to remove personal data before loading.

HOW DOES IT WORK?

- ETL: transform in a separate engine, then load clean tables.
- ELT: load raw tables first (for example in a raw schema), then SQL models build the clean layers.
- Keep the raw data so models can be rebuilt.

PRACTICAL EXAMPLE

> Raw Shopify orders load into a raw schema; a dbt model builds a cleaned orders table and a daily revenue table.

TOOL OR SOFTWARE

dbt, SQL

HANDS-ON EXERCISE

Describe for one business report whether you would use ETL or ELT and why, in five sentences.

COMMON MISTAKES

Transforming in place and losing the raw data needed to fix mistakes later.

INTERVIEW QUESTION

When would you choose ETL over ELT?

OFFICIAL AND FURTHER READING

- dbt: what is dbt (official)

A1.5

Pipeline reliability: idempotency, retries, incremental loads

WHAT IS IT?

> An idempotent pipeline gives the same result when run twice. Incremental loads process only new or changed data. Retries repeat a failed step with a wait between attempts.

WHY IS IT IMPORTANT?

> Pipelines fail and are re-run. Without idempotency, a re-run creates duplicate sales and wrong totals.

WHERE IS IT USED IN INDUSTRY?

Every production pipeline in finance, e-commerce and logistics must survive reruns and partial failures.

HOW DOES IT WORK?

- Use MERGE or upsert keyed on a business key instead of plain INSERT.
- Load into a staging table, validate, then swap or merge.
- Store a run ID and watermark so reruns know where they stopped.
- Retry transient errors (timeouts) with exponential backoff; fail fast on bad data.

PRACTICAL EXAMPLE

> A nightly job merges orders on order_id; running it twice still leaves one row per order.

TOOL OR SOFTWARE

SQL MERGE or INSERT ... ON CONFLICT (PostgreSQL)

HANDS-ON EXERCISE

Write a MERGE that upserts 1,000 orders twice and prove the row count is unchanged.

COMMON MISTAKES

Using INSERT for a daily load, so reruns duplicate rows.

INTERVIEW QUESTION

What does idempotent mean for a data pipeline?

OFFICIAL AND FURTHER READING

- PostgreSQL INSERT ... ON CONFLICT (official)

A1.6

Data modelling: conceptual, logical, physical; star and snowflake schemas

WHAT IS IT?

> Conceptual models name the business entities; logical models add attributes and keys; physical models are the actual tables. A star schema has one fact table of measures and dimension tables describing them.

WHY IS IT IMPORTANT?

> A clear model makes queries simple and consistent, so two teams get the same revenue number.

WHERE IS IT USED IN INDUSTRY?

Almost every BI platform (Power BI, Tableau, Looker) works best on a star schema.

HOW DOES IT WORK?

- Fact table: sales lines with amount, quantity and foreign keys.
- Dimension tables: customer, product, date, store.
- Snowflake schema normalises dimensions further (more joins, less duplication).

PRACTICAL EXAMPLE

> fact_sales(order_line_id, date_key, customer_key, product_key, amount) joined to dim_customer, dim_product, dim_date.

TOOL OR SOFTWARE

SQL, Power BI model view

HANDS-ON EXERCISE

Design a star schema for a coffee-shop chain with sales, products, stores and dates; list the grain of the fact table.

COMMON MISTAKES

Mixing grains in one fact table (order-level and line-level measures together).

INTERVIEW QUESTION

What is the grain of a fact table and why does it matter?

OFFICIAL AND FURTHER READING

- Kimball Group: dimensional modelling techniques

A1.7

Slowly Changing Dimensions (SCD) types 1, 2 and 3

WHAT IS IT?

> A slowly changing dimension is an attribute that changes over time, such as a customer's city. Type 1 overwrites it; Type 2 keeps history with valid-from and valid-to dates; Type 3 keeps one previous value.

WHY IS IT IMPORTANT?

> Reports about past periods must use the value that was true then. Type 2 makes that possible.

WHERE IS IT USED IN INDUSTRY?

Customer, employee and product history in banks, HR systems and retailers depend on SCD Type 2.

HOW DOES IT WORK?

- Type 1: update the row; history is lost.
- Type 2: close the old row (valid_to) and insert a new row with a new surrogate key.
- Type 3: add a column for the previous value.

PRACTICAL EXAMPLE

> A customer moves from Pune to Bengaluru on 1 Jan; the old row ends on 31 Dec and the new row starts on 1 Jan, so last year's sales still show Pune.

TOOL OR SOFTWARE

SQL

HANDS-ON EXERCISE

Write a Type 2 update for one customer change and show the two rows.

COMMON MISTAKES

Using Type 1 for attributes that reports need historically.

INTERVIEW QUESTION

What is the difference between SCD Type 1 and Type 2?

OFFICIAL AND FURTHER READING

- Kimball Group: slowly changing dimensions

A2 Data Warehousing, Lakes and Lakehouses

A2.1

Data warehouse, data mart, data lake and lakehouse

WHAT IS IT?

> A data warehouse stores cleaned, structured data for reporting. A data mart is a warehouse slice for one department. A data lake stores raw files of any type cheaply. A lakehouse adds warehouse features (tables, ACID transactions, SQL) on top of lake storage.

WHY IS IT IMPORTANT?

> Choosing the right storage decides cost, speed and who can use the data.

WHERE IS IT USED IN INDUSTRY?

Retailers and banks typically run a warehouse (Snowflake, BigQuery, Redshift); AI teams often use a lake or lakehouse for unstructured data.

HOW DOES IT WORK?

- Warehouse: schema on write, optimised for SQL.
- Lake: schema on read, cheap object storage (S3, Azure Blob, Google Cloud Storage).
- Lakehouse: table formats (Delta Lake, Apache Iceberg) give transactions on lake files.

PRACTICAL EXAMPLE

> Raw clickstream JSON sits in the lake; cleaned sessions go to the warehouse for BI.

TOOL OR SOFTWARE

BigQuery, Snowflake, Databricks (concepts)

HANDS-ON EXERCISE

Write a one-page comparison of warehouse, lake and lakehouse for a company with 3 TB of logs and a BI team.

COMMON MISTAKES

Putting raw logs in the warehouse without cost controls.

INTERVIEW QUESTION

What is the difference between a data lake and a data warehouse?

OFFICIAL AND FURTHER READING

- Snowflake: what is a data warehouse (official)
- Databricks: lakehouse (official)

A2.2

Medallion architecture: bronze, silver and gold

WHAT IS IT?

> Medallion architecture organises a lakehouse or warehouse in three layers: bronze (raw, as received), silver (cleaned and conformed), gold (business-ready tables and metrics).

WHY IS IT IMPORTANT?

> Each layer has one job, so bugs are found and fixed in one place, and analysts use only trusted gold tables.

WHERE IS IT USED IN INDUSTRY?

Used widely on Databricks, Microsoft Fabric and by dbt-based teams.

HOW DOES IT WORK?

- Bronze: append raw data with load time and source.
- Silver: deduplicate, fix types, apply keys and quality rules.
- Gold: aggregates and star schemas for dashboards and models.

PRACTICAL EXAMPLE

> bronze.orders (raw JSON) → silver.orders (typed, deduplicated) → gold.daily_revenue (used by Power BI).

TOOL OR SOFTWARE

dbt, Databricks, Fabric (concepts)

HANDS-ON EXERCISE

Sketch the three layers for an invoices dataset and name one quality check at each layer.

COMMON MISTAKES

Letting analysts query bronze tables that are not validated.

INTERVIEW QUESTION

Explain the medallion architecture.

OFFICIAL AND FURTHER READING

- Databricks: medallion architecture (official)

A3 Orchestration and Pipelines

A3.1

Apache Airflow: DAGs, tasks, scheduling and retries

WHAT IS IT?

> Airflow runs workflows as DAGs (directed acyclic graphs): tasks with dependencies, scheduled and retried, with logs and a web UI.

WHY IS IT IMPORTANT?

> Pipelines have many steps that must run in order and recover from failure. Orchestrators make that visible and repeatable.

WHERE IS IT USED IN INDUSTRY?

Common in data platforms at banks, media and e-commerce companies.

HOW DOES IT WORK?

- Define a DAG in Python with a schedule.
- Each task is an operator (Python, SQL, Bash, cloud service).
- Dependencies are set with >> (for example extract >> transform >> load).
- Retries and alerts are set per task.

PRACTICAL EXAMPLE

> A DAG runs at 02:00: extract orders, run dbt models, run data-quality tests, and alert the team if a test fails.

TOOL OR SOFTWARE

Apache Airflow

HANDS-ON EXERCISE

Install Airflow locally with Docker and build a three-task DAG that prints dates and fails on purpose once.

COMMON MISTAKES

Putting heavy data processing inside the orchestrator instead of calling a warehouse or Spark.

INTERVIEW QUESTION

What is a DAG and why are retries set per task?

OFFICIAL AND FURTHER READING

- Apache Airflow documentation (official)
- Airflow Docker quickstart (official)

A3.2

dbt: models, tests, documentation and lineage

WHAT IS IT?

> dbt (data build tool) lets analysts write SQL SELECT statements as models; dbt builds them in the warehouse in order, runs tests (not null, unique, relationships) and generates documentation and lineage graphs.

WHY IS IT IMPORTANT?

> dbt brings software practices (version control, tests, docs) to SQL transformations.

WHERE IS IT USED IN INDUSTRY?

Standard in analytics engineering teams; listed in many data analyst and analytics engineer job descriptions.

HOW DOES IT WORK?

- models/*.sql: each file is a table or view.
- ref('model') sets dependencies.
- schema.yml defines tests and descriptions.
- dbt run builds, dbt test checks, dbt docs generate the site.

PRACTICAL EXAMPLE

> stg_orders.sql cleans raw orders; fct_orders.sql joins to customers; a test asserts order_id is unique.

TOOL OR SOFTWARE

dbt Core (open source) with DuckDB or PostgreSQL

HANDS-ON EXERCISE

Build two dbt models over a sample orders table and add three tests.

COMMON MISTAKES

Writing one giant model instead of layers (staging, intermediate, marts).

INTERVIEW QUESTION

What do dbt tests check?

OFFICIAL AND FURTHER READING

- dbt documentation (official)
- dbt tutorial (official)

A3.3

Apache Spark and PySpark (big data processing)

WHAT IS IT?

> Spark processes large datasets in parallel across a cluster. PySpark is its Python API using DataFrames.

WHY IS IT IMPORTANT?

> Single machines run out of memory on large data; Spark splits the work and combines results.

WHERE IS IT USED IN INDUSTRY?

Used for large transformations in telecom, retail and AI feature pipelines.

HOW DOES IT WORK?

- DataFrames are distributed tables; transformations are lazy until an action (write, show).
- Partitioning splits data for parallel work; shuffles (joins, groupBy) move data between machines and are the costly step.

PRACTICAL EXAMPLE

```
from pyspark.sql import functions as F; df.groupBy('city').agg(F.sum('amount')).write.parquet(...)
```

TOOL OR SOFTWARE

PySpark (local mode), Databricks Community Edition

HANDS-ON EXERCISE

Run a groupBy on a 10-million-row file locally in PySpark and explain where the shuffle happens.

COMMON MISTAKES

Calling collect() on big data, which pulls everything to the driver.

INTERVIEW QUESTION

What is a shuffle in Spark and why is it expensive?

OFFICIAL AND FURTHER READING

- Apache Spark SQL guide (official)

A3.4

Streaming with Apache Kafka

WHAT IS IT?

> Kafka is a distributed log: producers write events to topics; consumers read them, at their own pace, from partitions.

WHY IS IT IMPORTANT?

> Real-time features (fraud alerts, stock levels, live dashboards) need events processed as they arrive.

WHERE IS IT USED IN INDUSTRY?

Used by banks, logistics and large e-commerce platforms.

HOW DOES IT WORK?

- Topic: a named stream; partitions give parallelism.
- Producer writes; consumer group reads and tracks offsets.
- Retention keeps events for replay.

PRACTICAL EXAMPLE

> Order events go to topic orders; a consumer updates the live inventory table.

TOOL OR SOFTWARE

Kafka (local Docker), Python kafka client

HANDS-ON EXERCISE

Produce 100 events and consume them in a script; then restart the consumer and show it resumes from its offset.

COMMON MISTAKES

Assuming message order across partitions, or not handling duplicate events.

INTERVIEW QUESTION

What is a Kafka partition used for?

OFFICIAL AND FURTHER READING

- Apache Kafka documentation (official)

A4 Data Quality, Observability and Governance Basics

A4.1

Data quality dimensions and rules

WHAT IS IT?

> Data quality is measured on accuracy, completeness, consistency, validity, uniqueness and timeliness. Rules turn these into checks that run on every load.

WHY IS IT IMPORTANT?

> Bad data makes reports wrong; checks catch it before people see it.

WHERE IS IT USED IN INDUSTRY?

Mandatory in finance, healthcare and regulated reporting.

HOW DOES IT WORK?

- Completeness: no missing mandatory fields.
- Uniqueness: one row per business key.
- Validity: values in range and in allowed lists.
- Timeliness: data arrives on schedule.
- Reconciliation: totals match the source.

PRACTICAL EXAMPLE

> Test: order_amount >= 0; test: customer_id exists in customers; test: daily row count within 20% of last week.

TOOL OR SOFTWARE

SQL checks, dbt tests, Great Expectations (open source)

HANDS-ON EXERCISE

Write ten SQL checks for a sales table and run them against a sample with planted errors.

COMMON MISTAKES

Checking quality only at the end of the pipeline, after bad data has spread.

INTERVIEW QUESTION

Name the six dimensions of data quality and give an example of each.

OFFICIAL AND FURTHER READING

- Great Expectations docs (official)
- dbt tests (official)

A4.2

Data observability and pipeline monitoring

WHAT IS IT?

> Data observability tracks freshness, volume, schema changes and distribution of data, and alerts when something looks wrong.

WHY IS IT IMPORTANT?

> Pipelines can succeed technically and still load empty or wrong data. Observability catches that.

WHERE IS IT USED IN INDUSTRY?

Data platform teams use freshness and volume alerts on critical tables.

HOW DOES IT WORK?

- Freshness: last update time against an agreed SLA.
- Volume: row count compared to history.
- Schema: alerts when a column is added, removed or changes type.
- Alerts go to the owner with a link to the failed run.

PRACTICAL EXAMPLE

> The orders table has no new rows by 08:00; an alert opens an incident for the data owner.

TOOL OR SOFTWARE

Airflow alerts, dbt source freshness

HANDS-ON EXERCISE

Set a freshness check on a sample table and trigger it by stopping the load.

COMMON MISTAKES

Only monitoring whether the job ran, not whether the data is right.

INTERVIEW QUESTION

What is data freshness and how would you monitor it?

OFFICIAL AND FURTHER READING

- dbt source freshness (official)

A4.3

Data catalogues, lineage and classification

WHAT IS IT?

> A data catalogue lists datasets with owners and descriptions. Lineage shows where data came from and where it goes. Classification labels data as public, internal, confidential or restricted.

WHY IS IT IMPORTANT?

> People can find trusted data, and impact of a change is known before it breaks reports.

WHERE IS IT USED IN INDUSTRY?

Enterprises use catalogues such as Microsoft Purview, Collibra and Atlan.

HOW DOES IT WORK?

- Register each table with owner, description and tags.
- Lineage graphs come from dbt or the catalogue.
- Classification drives access and retention rules.

PRACTICAL EXAMPLE

> A table tagged PII restricted to HR analysts; lineage shows it feeds a payroll dashboard.

TOOL OR SOFTWARE

dbt docs (lineage), Microsoft Purview (overview)

HANDS-ON EXERCISE

Document one dbt project with descriptions and a lineage graph screenshot.

COMMON MISTAKES

Having no owner for a dataset, so nobody fixes it.

INTERVIEW QUESTION

What is data lineage and why does it matter?

OFFICIAL AND FURTHER READING

- Microsoft Purview documentation (official)

A9 Cloud Fundamentals for Data

A9.1

Cloud basics: IaaS, PaaS, SaaS, regions and identity

WHAT IS IT?

> Cloud computing rents compute, storage and services over the internet. IaaS gives virtual machines; PaaS gives managed platforms; SaaS gives finished applications. Regions and availability zones decide where data lives. Identity and access management (IAM) decides who can do what.

WHY IS IT IMPORTANT?

> Data platforms run in the cloud; wrong permissions or regions cause security and compliance failures.

WHERE IS IT USED IN INDUSTRY?

Almost every data role uses cloud storage, compute and IAM daily.

HOW DOES IT WORK?

- Choose a region close to users and matching data residency rules.
- Give services the least privilege they need (roles, not shared keys).
- Use managed services to reduce operations work.

PRACTICAL EXAMPLE

> A data pipeline runs as a service role that can read one bucket and write to one warehouse schema, nothing else.

TOOL OR SOFTWARE

AWS, Azure and Google Cloud consoles

HANDS-ON EXERCISE

Draw the IAM roles for a pipeline that reads files from storage and writes to a warehouse.

COMMON MISTAKES

Using a root or owner account for daily work, or storing keys in code.

INTERVIEW QUESTION

What is least privilege and why does it matter for data pipelines?

OFFICIAL AND FURTHER READING

- AWS: what is cloud computing (official)
- Google Cloud documentation (official)

A9.2

Data services on AWS, Azure and Google Cloud (mapping)

WHAT IS IT?

> The three big clouds offer similar data services under different names. Knowing the mapping lets you move between them.

WHY IS IT IMPORTANT?

> Customers use different clouds; FDEs and data engineers must read any of them.

WHERE IS IT USED IN INDUSTRY?

Enterprise job descriptions name one or more of these clouds.

HOW DOES IT WORK?

- Object storage: AWS S3, Azure Blob Storage, Google Cloud Storage.
- Serverless ETL: AWS Glue, Azure Data Factory, Google Cloud Dataflow.
- Warehouse: Amazon Redshift, Azure Synapse, Google BigQuery.
- Streaming: Amazon Kinesis or MSK, Azure Event Hubs, Google Pub/Sub.
- Analytics platform: Databricks on all three, Microsoft Fabric on Azure, Snowflake on all three.

PRACTICAL EXAMPLE

> The same medallion pipeline runs on S3 + Glue + Redshift or on Blob + Data Factory + Synapse.

TOOL OR SOFTWARE

Cloud consoles and CLIs

HANDS-ON EXERCISE

Make a table with the service names for storage, ETL, warehouse, streaming and orchestration on each cloud.

COMMON MISTAKES

Assuming a service works the same in every cloud; limits and pricing differ.

INTERVIEW QUESTION

Which cloud data services would you use for a batch warehouse on Azure?

OFFICIAL AND FURTHER READING

- AWS data analytics services (official)
- Azure Data Factory docs (official)
- Google BigQuery docs (official)

A9.3

Cloud cost optimisation for data platforms

WHAT IS IT?

> Cost optimisation keeps cloud spend proportional to value: right-sized compute, auto-suspend, partitioning, storage tiers and budgets.

WHY IS IT IMPORTANT?

> Warehouses and Spark clusters left running cost thousands of rupees or dollars a month for nothing.

WHERE IS IT USED IN INDUSTRY?

Cloud cost is a daily concern for data platform teams and FDE customers.

HOW DOES IT WORK?

- Set budgets and alerts on day one.
- Auto-suspend idle warehouses and clusters.
- Partition and cluster big tables; avoid SELECT * on huge tables.
- Move cold data to cheaper storage tiers.

PRACTICAL EXAMPLE

> A warehouse set to auto-suspend after 5 minutes cuts a monthly bill by most of the idle time.

TOOL OR SOFTWARE

Cloud cost dashboards

HANDS-ON EXERCISE

Estimate the monthly cost of a pipeline that runs hourly for 2 hours and propose three cost cuts.

COMMON MISTAKES

Leaving dev clusters running overnight.

INTERVIEW QUESTION

How would you reduce a warehouse bill that doubled this month?

OFFICIAL AND FURTHER READING

- AWS cost optimization pillar (official)

A10 Microsoft Fabric and Databricks (Modern Data Platforms)

A10.1

Microsoft Fabric: lakehouse, warehouse, dataflows, semantic models

WHAT IS IT?

> Microsoft Fabric is an all-in-one analytics platform: OneLake storage, lakehouses and warehouses, pipelines, dataflows, notebooks and Power BI semantic models in one place.

WHY IS IT IMPORTANT?

> Many Power BI users now build their models on Fabric; knowing its parts helps with Power BI roles.

WHERE IS IT USED IN INDUSTRY?

Used by Microsoft-centric enterprises and Power BI consultancies.

HOW DOES IT WORK?

- OneLake: one logical data lake for the organisation.
- Lakehouse: Delta tables with SQL and Spark access.
- Warehouse: SQL warehouse.
- Semantic model: the Power BI dataset used by reports.
- Deployment pipelines move content from dev to test to prod.

PRACTICAL EXAMPLE

> A sales lakehouse feeds a semantic model; the Power BI report connects in Direct Lake mode.

TOOL OR SOFTWARE

Microsoft Fabric trial (free trial, check current terms)

HANDS-ON EXERCISE

Create a Fabric trial lakehouse, load a CSV and build a semantic model.

COMMON MISTAKES

Assuming trial limits apply in production; check licences and capacity.

INTERVIEW QUESTION

What is a semantic model in Power BI?

OFFICIAL AND FURTHER READING

- Microsoft Fabric documentation (official)

A10.2

Databricks: notebooks, Delta Lake, Spark jobs and Unity Catalog

WHAT IS IT?

> Databricks is a managed Spark platform with notebooks, Delta Lake tables (ACID transactions on files), jobs and Unity Catalog for governance.

WHY IS IT IMPORTANT?

> Large AI and data teams standardise on Databricks.

WHERE IS IT USED IN INDUSTRY?

Common in finance, healthcare and AI product companies.

HOW DOES IT WORK?

- Notebooks run PySpark and SQL.
- Delta tables support MERGE, time travel and schema evolution.
- Unity Catalog controls access and lineage.

PRACTICAL EXAMPLE

> A MERGE into a Delta silver table keeps the table current from a daily file drop.

TOOL OR SOFTWARE

Databricks Free Edition (check current availability)

HANDS-ON EXERCISE

Create a Delta table from a CSV and run a MERGE.

COMMON MISTAKES

Skipping access controls on shared catalogs.

INTERVIEW QUESTION

Why is Delta Lake useful for pipelines that need reruns?

OFFICIAL AND FURTHER READING

- Databricks documentation (official)

A11 Containers, Deployment and Operations

A11.1

Docker fundamentals for data work

WHAT IS IT?

> Docker packages an app and its dependencies into an image that runs the same everywhere. A container is a running image.

WHY IS IT IMPORTANT?

> Pipelines and notebooks must run the same on a laptop, in CI and in the cloud.

WHERE IS IT USED IN INDUSTRY?

Standard in data engineering and AI deployment.

HOW DOES IT WORK?

- Dockerfile lists the build steps.
- docker build creates an image; docker run starts a container.
- docker compose starts several services (database, Airflow, app) together.

PRACTICAL EXAMPLE

> A Dockerfile installs Python and dependencies, copies the pipeline code and runs it on a schedule.

TOOL OR SOFTWARE

Docker Desktop (check current licence terms for your use)

HANDS-ON EXERCISE

Containerise a small Python script that reads a CSV and writes Parquet.

COMMON MISTAKES

Baking secrets into the image.

INTERVIEW QUESTION

What is the difference between an image and a container?

OFFICIAL AND FURTHER READING

- Docker get started (official)

A11.2

Kubernetes basics for data platforms

WHAT IS IT?

> Kubernetes runs containers across a cluster: deployments keep a number of pods running, services route traffic, and jobs run tasks to completion.

WHY IS IT IMPORTANT?

> Large data and AI platforms run on Kubernetes; engineers need to read manifests and debug pods.

WHERE IS IT USED IN INDUSTRY?

Used by large enterprises on EKS, AKS and GKE.

HOW DOES IT WORK?

- Pod: one or more containers.
- Deployment: keeps N replicas.
- Service: stable network address.
- Job and CronJob: run batch work.

PRACTICAL EXAMPLE

> A CronJob runs the nightly pipeline container at 02:00 on the cluster.

TOOL OR SOFTWARE

kubectl and a local cluster (kind)

HANDS-ON EXERCISE

Deploy a CronJob that prints the date and check its logs.

COMMON MISTAKES

Running stateful databases without understanding storage.

INTERVIEW QUESTION

What is a Kubernetes CronJob used for?

OFFICIAL AND FURTHER READING

- Kubernetes concepts (official)

A11.3

CI/CD for data pipelines and dbt projects

WHAT IS IT?

> CI/CD runs tests and deploys changes automatically: on each pull request tests run, and on merge the pipeline deploys to the target environment.

WHY IS IT IMPORTANT?

> Data changes break reports; automated tests catch them before production.

WHERE IS IT USED IN INDUSTRY?

Standard for analytics engineering and data platform teams.

HOW DOES IT WORK?

- GitHub Actions workflow: lint, dbt build on a test schema, dbt test.
- Deploy on merge to main.
- Store secrets in the CI secret store, never in code.

PRACTICAL EXAMPLE

> A pull request runs dbt build against a test schema; merge deploys to production.

TOOL OR SOFTWARE

GitHub Actions

HANDS-ON EXERCISE

Write a workflow that runs dbt test on each pull request.

COMMON MISTAKES

Testing only in production.

INTERVIEW QUESTION

How would you add tests for a dbt project in CI?

OFFICIAL AND FURTHER READING

- GitHub Actions documentation (official)

A12 Data Architecture and System Design Basics

A12.1

Data architecture: sources, ingestion, storage, serving and governance

WHAT IS IT?

> A data architecture diagram shows each source, pipeline, store, consumer and control (security, quality, cost).

WHY IS IT IMPORTANT?

> Architects agree the design with stakeholders before building; diagrams make the design reviewable.

WHERE IS IT USED IN INDUSTRY?

Architects and FDEs present these diagrams to customers.

HOW DOES IT WORK?

- Draw sources on the left and consumers on the right.
- Label every pipeline with its frequency and owner.
- Mark where data is classified and where quality checks run.

PRACTICAL EXAMPLE

> A retail architecture: POS and web data → Kafka → lakehouse bronze/silver/gold → dashboards and an ML feature store.

TOOL OR SOFTWARE

draw.io, Excalidraw, Mermaid

HANDS-ON EXERCISE

Draw the data architecture for a clinic's appointments and billing, with one quality check at each layer.

COMMON MISTAKES

Drawing tools without showing data ownership and security.

INTERVIEW QUESTION

How would you present a data architecture to a CTO?

OFFICIAL AND FURTHER READING

- C4 model for diagrams (official site)

A12.2

System design fundamentals for data and AI systems

WHAT IS IT?

System design picks components and trade-offs for scale, latency, reliability and cost. The usual steps: clarify requirements, estimate load, design the flow, choose storage, plan failure handling.

WHY IS IT IMPORTANT?

Interviews for data engineers and FDE roles use system design questions heavily.

WHERE IS IT USED IN INDUSTRY?

Common in technical rounds at data platform and AI companies.

HOW DOES IT WORK?

- Clarify users, data volume, freshness and latency.
- Estimate rows per day and storage per year.
- Choose batch or streaming; choose storage by access pattern.
- Plan retries, monitoring and backfills.

PRACTICAL EXAMPLE

Design a daily pipeline for 50 million events: batch versus streaming, storage format, partitioning by date, retries and backfills.

TOOL OR SOFTWARE

Whiteboard or document

HANDS-ON EXERCISE

Write a one-page design for a real-time fraud feature pipeline with trade-offs.

COMMON MISTAKES

Jumping into tools before agreeing requirements.

INTERVIEW QUESTION

How do you design a pipeline that must handle reprocessing of last year's data?

OFFICIAL AND FURTHER READING

- System Design Primer (open source)

A13 Hands-on Cloud Lab (Module 10)

A13.1

Lab: load, query and share data in a cloud warehouse

WHAT IS IT?

> A guided lab that takes one CSV file from your laptop to a cloud warehouse, queries it with SQL and connects a dashboard, then cleans everything up.

WHY IS IT IMPORTANT?

> Reading about cloud services is not the same as using them. One small end-to-end lab builds the confidence interviewers look for.

WHERE IS IT USED IN INDUSTRY?

Every cloud data team does this daily: land a file, model it, query it and serve it to a dashboard.

HOW DOES IT WORK?

- Create a free account or sandbox. Google BigQuery offers a sandbox without a payment card; Azure and AWS have free tiers. Check the current limits before you start.
- Create a dataset (BigQuery) or database (Azure SQL, Amazon Redshift Serverless or Athena) in one region.
- Upload a public CSV (for example a sales or taxi trips sample) and let the tool detect the schema; fix any wrong column types.
- Write three queries: total by month, top 10 by revenue, and a window-function ranking. Save one as a view.
- Connect Looker Studio, Power BI or Metabase to the view and build one chart.
- Delete the dataset, tables and any storage buckets; confirm the billing page shows no running services.

PRACTICAL EXAMPLE

> A 50,000-row orders CSV becomes a BigQuery table; a view gives revenue by month; Looker Studio shows a line chart; the dataset is deleted at the end.

TOOL OR SOFTWARE

Google BigQuery sandbox, Looker Studio; or Azure SQL / Amazon Athena with Power BI

HANDS-ON EXERCISE

Complete the six steps, take a screenshot of each, and write a five-line README with the cost (should be zero) and what you would change for production.

COMMON MISTAKES

Leaving resources running after a lab, or uploading real customer data to a personal account.

INTERVIEW QUESTION

Walk me through how you would land a CSV in a cloud warehouse and make it available to a dashboard.

OFFICIAL AND FURTHER READING

- BigQuery sandbox (official)
- Azure free account (official)

- AWS Free Tier (official)

A13.2

Cost guardrails for cloud labs and small projects

WHAT IS IT?

> Settings and habits that stop a learning project from creating an unexpected bill.

WHY IS IT IMPORTANT?

> Cloud services charge by use; a forgotten cluster or an unbounded query can cost real money.

WHERE IS IT USED IN INDUSTRY?

Teams set budgets and alerts on every account; learners should do the same on day one.

HOW DOES IT WORK?

- Set a budget (for example ₹500) with email alerts at 50%, 80% and 100%.
- Prefer serverless and sandbox options that cost nothing when idle.
- In warehouses, select only needed columns and filter on partitioned dates to scan less data.
- Tag every resource with project and owner; delete by tag at the end.
- Review the billing page weekly while learning.

PRACTICAL EXAMPLE

> A learner sets a ₹500 budget alert, queries only three columns from a partitioned table and deletes the project after the capstone.

TOOL OR SOFTWARE

Cloud billing budgets and alerts; BigQuery query validator (shows bytes before running)

HANDS-ON EXERCISE

Set up a budget alert on your cloud account and screenshot it; run one query and note the bytes scanned before and after selecting fewer columns.

COMMON MISTAKES

Using SELECT * on large tables, or assuming free tiers never charge.

INTERVIEW QUESTION

How would you keep cloud costs under control for a new data project?

OFFICIAL AND FURTHER READING

- Google Cloud budgets (official)
- AWS Budgets (official)